

1. A particle of mass 2 kg is moving with velocity $(5\mathbf{i} + 3\mathbf{j})\text{ m s}^{-1}$ when it receives an impulse $\mathbf{I}\text{ N s}$, such that $\mathbf{I} = a\mathbf{i} + b\mathbf{j}$

Immediately after receiving the impulse, the particle is moving with velocity $\lambda(\mathbf{i} + \mathbf{j})\text{ m s}^{-1}$, where λ is a constant.

Given that the magnitude of $\mathbf{I}$ is $\sqrt{40}$, find the two possible impulses.

(5)

2. A truck of weight 9000 N is travelling up a hill on a straight road that is inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{15}$

When the truck travels up the hill with the engine working at $3P$ watts, the truck is moving at a constant speed of 12 m s^{-1}

Later on, the truck travels down the hill along the same road, with the engine working at P watts. At the instant when the speed of the truck is 12 m s^{-1} , the acceleration of the truck is $\frac{g}{20}$

The resistance to motion of the truck from non-gravitational forces is a constant force of magnitude R newtons in all circumstances.

Find (i) the value of P ,

(ii) the value of R .

(9)

3.

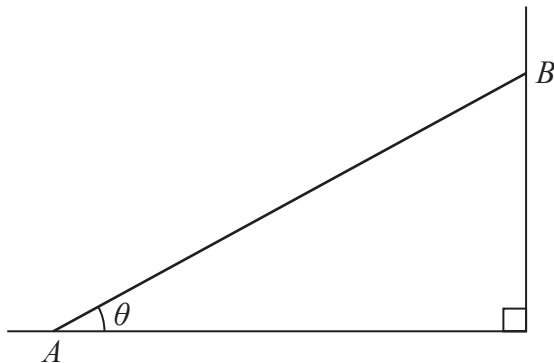


Figure 1

A uniform rod AB , of mass 25 kg and length 3 m, has end A resting on rough horizontal ground. The end B rests against a rough vertical wall.

The rod is in a vertical plane perpendicular to the wall.

The coefficient of friction between the rod and the ground is $\frac{4}{5}$

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The rod rests in limiting equilibrium.

The rod is at an angle of θ to the ground, as shown in Figure 1.

Find the exact value of $\tan \theta$.

4.

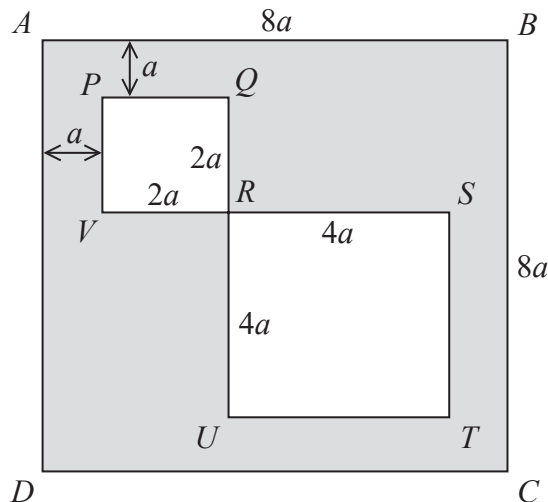


Figure 2

The uniform lamina L , shown shaded in Figure 2, is formed by removing the square $PQRV$, of side $2a$, and the square $RSTU$, of side $4a$, from a uniform square lamina $ABCD$, of side $8a$. The lines QRU and VRS are straight. The side AD is parallel to PV and the side AB is parallel to PQ . The distance between AD and PV is a and the distance between AB and PQ is a . The centre of mass of L is at the point G .

- (a) Show that the distance of G from the side AD is $\frac{42}{11}a$ (5)

The mass of L is M . A particle of mass kM is attached to L at C .

The lamina, with the attached particle, is freely suspended from B and hangs in equilibrium with BC making an angle of 45° with the horizontal.

- (b) Find the value of k .

(4)

5. At time t seconds ($t \geq 0$), a particle P has velocity $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = (3t^2 - 9t + 6)\mathbf{i} + (t^2 + t - 6)\mathbf{j}$$

- (a) Find the acceleration of P when $t = 3$

(3)

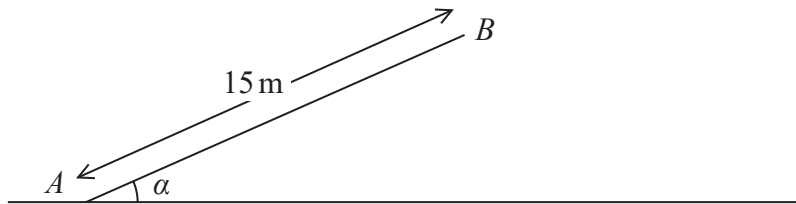
When $t = 0$, P is at the fixed point O .

The particle comes to instantaneous rest at the point A .

- (b) Find the distance OA .

(7)

6.

**Figure 3**

A rough straight ramp is fixed to horizontal ground. The ramp has length 15 m and is inclined at an angle α to the ground, where $\tan \alpha = \frac{5}{12}$. The line AB is a line of greatest slope of the ramp, where A is at the bottom of the ramp, and B is at the top of the ramp, as shown in Figure 3.

A particle P of mass 6 kg is projected up the ramp with speed 14 m s^{-1} from A in a straight line towards B . The coefficient of friction between P and the ramp is 0.25

(a) Find the work done against friction as P moves from A to B .

(3)

At the instant P reaches B , the speed of P is $v \text{ m s}^{-1}$. After leaving the ramp at B , the particle P moves freely under gravity until it hits the horizontal ground at the point C . Immediately before hitting the ground at C , the speed of P is $w \text{ m s}^{-1}$

(b) Use the work-energy principle to find

(i) the value of v ,

(ii) the value of w .

(7)

7. Particle A of mass $3m$ is moving in a straight line with speed $2u$ on a smooth horizontal surface. Particle A collides directly with particle B of mass m , which is moving along the same straight line and in the same direction as A .

Immediately before the collision, the speed of B is u .

As a result of the collision, the direction of motion of B is unchanged and the kinetic energy gained by B is $\frac{48}{25}mu^2$

- (a) Find the coefficient of restitution between A and B .

(8)

After the collision, B hits a smooth fixed vertical wall that is perpendicular to the direction of motion of B . The coefficient of restitution between B and the wall is f .

Given that the speed of B immediately after first hitting the wall is equal to the speed of A immediately after its first collision with B ,

- (b) find the value of f .

(2)

8. [In this question, the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, with $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.]

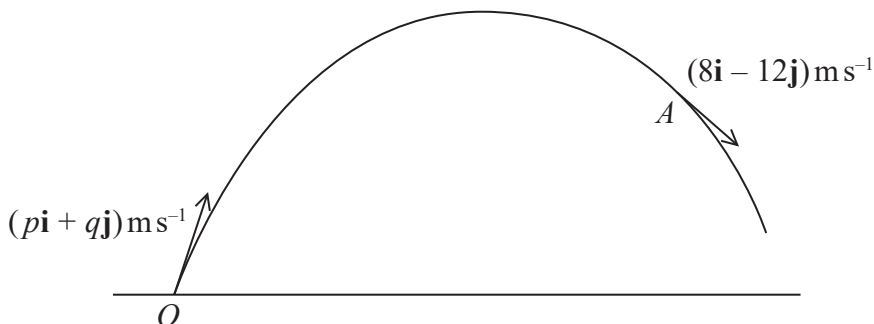


Figure 4

At time $t = 0$, a small ball is projected from a fixed point O on horizontal ground. The ball is projected from O with velocity $(p\mathbf{i} + q\mathbf{j}) \text{ m s}^{-1}$, where p and q are positive constants. The ball moves freely under gravity.

At time $t = 3$ seconds, the ball passes through the point A with velocity $(8\mathbf{i} - 12\mathbf{j}) \text{ m s}^{-1}$, as shown in Figure 4.

- (a) Find the speed of the ball at the instant it is projected from O . (5)

For an interval of T seconds the speed, $v \text{ m s}^{-1}$, of the ball is such that $v \leq 10$

- (b) Find the value of T . (4)

At the point B on the path of the ball, the direction of motion of the ball is perpendicular to the direction of motion of the ball at A .

- (c) Find the vertical height of B above A . (4)
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